6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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Abstract.

Understanding climate change relies heavily on grasping the role of cloud radiative effect and its climate feedback. However, these factors pose the greatest uncertainty among atmospheric processes. This uncertainty stems partly from the limited global-scale cloud measurements. To address this gap, our study presents a six-year statistical analysis of cloud coverage and microphysics properties at the ACTRIS-CCRES AGORA observatory in Granada, Spain. Here, we utilized CLOUDNET post-processed data (e.g target classification, liquid water path, droplet effective radius and so on) of remote sensing instruments and microwave radiometer thermodynamic profiles. The analysis focuses on single layer cloud properties comparison for liquid, ice, mixed phase, liquid with rain, and mixed phase with rain clouds. Therefore, a distinct seasonal pattern in cloud occurrence was observed. During the summer (June-August), single layer clouds occurrence is less than 10%, and precipitation is absent. Conversely, cloud occurrence significantly increases during winter (December-February) and spring (March-May), where Ice and mixed phase clouds with rain are the predominant type. Additionally, a negligible presence of liquid clouds with rain were found. Seasonal trends in liquid water content (LWC) and droplet effective radius are unclear for mixed-phase and liquid clouds. Effective radius remains stable year-round, around 5 μm for liquid clouds below 3 km, while for mixed-phase clouds it can vary between 5 and 12 μm, mainly found at 1-2 km and 5-9 km. LWC shows higher values in cold periods and lower in summer, but with significant variability across both cloud types. Same seasonal patterns for ice water content (IWC) in ice and mixed-phase clouds were challenging to identify. Monthly profiles showed little variation, although values tended to rise in summer. Ice clouds typically have effective radius between 20-30 μm at 8-12 km altitude, peaking at 65 μm in summer. Mixed-phase clouds exhibit ice effective radius around 50 μm between 1-7 km altitude. The mean geometric thickness of ice and mixed-phase clouds throughout the seasons is 1.2 $\pm$ 1.3 km and 2.0 $\pm$ 1.6 km, respectively, contrasting sharply with the 0.2 $\pm$ 0.2 km observed in liquid clouds. The monthly variability in thickness is substantial enough to obscure any seasonal differences. Similarly, the analysis of cloud base height reveals mean values of 1.9 $\pm$ 2.0 km (excluding summer) for liquid clouds, 4.8 $\pm$ 2.2 km for mixed-phase clouds, and 7.0 $\pm$ 2.4 km for ice clouds. During summer a different vertical distribution of cloud base was found due to different mechanisms of cloud formation. Conclusively, a statistical analysis was conducted on single-layer cloud properties in the Southeast of Spain, highlighting the mainly cloud type associated with rainfall in the region.

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1 Introduction

Clouds are one of the major atmospheric components driving the earth's radiative budget and hydrological cycle variability. Since they roughly cover 2/3 of the Earth's surface **?**, their physical properties play an important role in climate feedbacks such as surface temperature change rate. In particular, clouds are responsible for precipitation patterns and can contribute to increased surface temperature by trapping thermal radiation in the atmosphere **?**. Given that, raised temperatures and water shortage accounts for more severe drought stress in mediterranean regions **??**. In addition, the fifth assessment report **?** suggests the Mediterranean basin will suffer a reduction of precipitation **??**. In general, this area is known for experiencing frequent severe weather events and significant climate variability **??**, being of particular scientific and societal interest. However, surface temperature feedback due cloud radiative effect is still poorly understood **?**. Moreover, further projections using GCMs (global climate models) are still challenging due to the difficulty in partitioning liquid and ice content for mixed phase clouds **?**. Particularly, assimilation of cloud types with precipitation showed the importance of ice process in driving precipitation for warm clouds Bühl et al. (2016); **?**. From this perspective, measurements of cloud doppler radar can be used to retrieve droplets and ice microphysics properties such as effective radius, liquid water (LWC) content and ice water content (IWC) **???**. In combination with a ceilometer, they can also determine cloud boundaries, i.e. base and tops. Thus, in order to understand seasonal trends on cloud occurrence and their physical properties, long-term measurements of remote sensing instruments are necessary. Therefore, a statistical description of cloud type occurrences in southeast Spain using passive and active remote sensing measurements to comprehend cloud and precipitation patterns in the Mediterranean basin was achieved.

The presented study used more than half a decade of Cloudnet post-processed data **?** and microwave radiometer profiles at the Spanish AGORA-CCRES station (Andalusian Global ObseRvatory of the Atmosphere - Centre of Cloud Remote Sensing). The AGORA station has been a part of the Cloudnet since yearXXX, and was integrated into ACTRIS CCRES in yearXXX as its new facility. Their products have been continuously determined using a synergistic approach of cloud observations using a 94-GHz doppler radar, with multiple scanning capabilities, a microwave radiometer and a CHM15k ceilometer. Here, Cloudnet target classification is employed to cloud type classification as **??**. This product also allowed the partitioning of retrieved droplets and ice microphysical properties within each cloud type. Thus, an inter-annual statistical analysis on cloud occurrence and microphysical properties is presented.

Cloudnet products have been explored in many recent studies. **?** used this dataset at German Alps to analyze cloud variability at a particular high-altitude location. **?** studied data from the Leipzig observatory LACROS (Leipzig Aerosol and Cloud Remote Observations System), to evaluate ice process for mixed phase clouds. **??** focused their study on cloud statistics properties for different types of clouds in the Arctic and the Bucharest-Magurele site, respectively. Differently from the previously cited studies, our station has particularly thermodynamic conditions, with elevated temperatures and less humidity. Besides that, we also described microphysical properties for different cloud types, which can be extremely useful for feeding radiative transfer models in order to estimate cloud radiative effects. **?** and **?** used microphysics estimation from XXX and XXX to perform numerical estimation of cloud radiative effect using the libRadtran tool.

The paper is organized as follows. Sec **??** describes the dataset and the methods used for cloud classification. Sec **??** first describes the cloud correlation with thermodynamic conditions and then evaluates the inter-annual cloud properties variability. Finally, SecXXX presents the concluding remarks and contextualize the main findings within the climate research topic.

2 Data and Methods

2.1 Site and database

An extensive dataset ranging the years 2018 to 2023 was used. In particular, Cloudnet (Illingworth et al., 2007) post-processed high level data and microwave radiometer humidity and temperature profiles. Concerning the 6 years of cloudnet dataset, Figure 1 shows the monthly data availability. These products were inferred by measurements carried out with a set of ground-based remote sensing instruments - managed at the ACTRIS-CCRES Granada site facility (CCRES stands for Centre for Cloud Remote Sensing) at 37°2E, 3.6°S and 680 m. This station is as part of the University of Granada (UGR) and it is integrated into the AGORA observatory. It is located within a broad intramontane depression in the foothills of the tallest mountain massif of the Iberian Peninsula, Sierra Nevada.

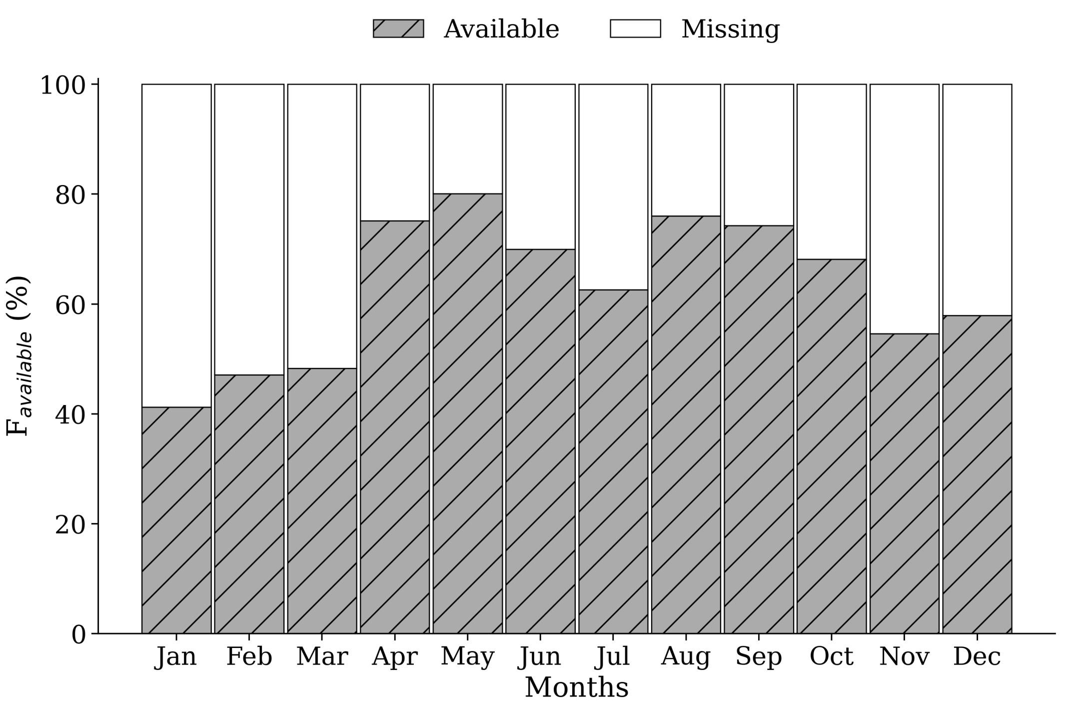

Figure 1. ...

Table 1. This is a table with some data.

Instrument	Bands / wavelenghts	Measured	Products	Range	Time	References
94 GHz Cloud Doppler Radar	94 GHz (W-band)	Doppler velocity spectrum (DVS)	ν_D , Z, S_w	[13,51] m[a]	[1.1, 3.6] s[b]	Kuchler, 2017
MWR RPG	22-31 GHz (K-bad)	Brightness	q and T	[10, 300] m[c]	[2, 2.5] min	Navas-Guzmán et al., 2014
HATPRO	51-58 GHz (V-band)	Temperature	LWP	-	2 s	
CHM15k Nimbus Ceilometer	1064 nm		β	15 m	15 s	Cazorla, et al., 2017

The Cloudnet processed data was determined by a synergy between a RPG-HATPRO radiometer, a CHM15k ceilometer, a RPG-FMCW 94 GHz cloud Doppler radar and the European model ECMWF (an acronym for European Centre for Medium-Range Weather Forecasts). These instruments have been continuously operated since 2018, and their data has been processed by the CloudnetPy (Illingworth et al., 2007), performing reflectivity attenuation correction, error estimation and quality flag assignment. Particularly, attenuation corrections assists in prevention misclassification of atmospheric elements through liquid water content (LWC) estimation. This variable is obtained by cloud boundaries provided by lidar and radar synergy, and microwave radiometer liquid water path (LWP) **?**. Thus, uncertainties retrieving LWC during a raining event can be high enough to frustrate attenuation corrections.

CLOUDNET processed data were sampled in a common grid with time resolution of 30 s, and height bins according to cloud radar range resolution, which differ according to radar chirp table configuration. This resolution vary with altitude and underwent some changes throughout the measurement period. Consequently, to ensure consistency in vertical resolution of the data, a regridding process was implemented to facilitate monthly statistical analysis when needed. The regriding consisted to change altitude resolution to 60 m. Table **??** associate bin sizes with their respective height intervals. This table also gives operational information of instruments at Granada site and their main post-processed variables. Differently, microwave radiometer humidity and temperature profiles were not processed by CloudnetPy, but were provided by its own manufacturing software. High level data included in the analysis were target classification, liquid water path (LWP) ice water path (IWP), droplet effective radius, and ice effective radius, where target classification apportioned to classify cloud layers as explained in section 2.3. The classification method was adapted from Pîrloagă et al. (2022) and Nomokonova et al. (2019).

2.2 Cloudnet particle classification scheme

Target classification product assigns an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Each number are related to atmospheric elements in the following order: clear sky, droplets, drizzle or rain, drizzle & droplets, ice, ice &

[a]It depends on the chirp type

[b]It also depends on the chirp type

[c]Humidity and temperature profiles has a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300m between 4.4 and 10 km

Cloudnet products	Input role	σ_{re}/r_e	References
Liquid water content (LWC)	Z and β: cloub base and top identification T_{ECMWF}, P_{ECMWF}: adiabatic LWC MWR or CDR LWP to scale LWC	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T_{ECMWF} profiles	30-40%	Hogan et al. (2006) Delanoë et al. 2007
Droplet effetive radius ($r_{e,liq}$)	Z, and N and σ of lognormal PSD	15%	Frisch et al. (2002)
Ice effective radius ($r_{e,ice}$)	Z and T_{ECMWF}	50%	Griesche et al.(2020)

droplets, melting ice, melting & droplets, aerosol, insects, aerosol & insect. These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles is used to establish warm and cold regions to classify liquid and ice hydrometeors. Further, liquid droplets boundaries were resolved with lidar attenuated backscatter coefficient (β), and radar reflectivity. The first is sensible to small liquid droplets present in cloud base, as the second are able to penetrate the cloud and detect cloud tops. In addition, liquid layers respect a threshold of T < -40 °C, since it is not possible to keep water in liquid phase bellow that temperature. Hence, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echos analysis once particles boundary were already determined. Therefore, falling cold particle were just classified as ice, without distinguishing ice from precipitating ice. Next, melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on Hogan and O'connor (2004).

2.3 Assortment algorithm

The assortment algoritm classifies groups of hydrometeors and cloud layers. In terms of hydrometeors, this algorithm generate masks of hydrometeor occurrence for 3 groups: liquid, ice and "total", where liquid class encompass all possible liquid pixels, which are "droplets", "drizzle or rain" and "drizzle & droplets". Ice encompass only "ice" pixels, and total is all possible hydrometeors, which are "droplets", "drizzle or rain", "drizzle & droplets", "ice", "ice & droplets", "melting ice" and "melting & droplets".

Subsequently, cloud layers in atmospheric column were classified trough pixel sequence inquiry. These pixels sequences were divided in the following 5 groups according to Pîrloagă et al. (2022): liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only "ice" pixel sequence was encountered. As well as, mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Different from previous studies, a threshold of 20 bins (256-1022 m) bellow cloud was set to reject precipitation events, assuming that rain droplets are not falling from the actual cloud layer. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (Nomokonova et al., 2019) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighbourhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on Pîrloagă et al. (2022); Nomokonova et al. (2019), but theses values are not currently well defined, their determination has been guided by empirical experience.

3 Results and discussion

3.1 Thermodynamic influences on cloud frequency

It is widely recognized that thermodynamic conditions can affect cloud formation (**?**). In order to relate both, this section makes a comparison between monthly thermodynamic profiles and different hydrometeors and cloud type occurrence.

Temperature and humidity profiles used as baselines were those collected during clear sky condition, and are shown in Figure 2. This figure can easily identify cold and warm regimes, where cold regimes spans between winter and spring, while warm regimes spans between summer and autumn. This annual variability was also observed by Pîrloagă et al. (2022); Nomokonova et al. (2019), however, higher temperature values experienced by our station is only closely resembling those reported by Pîrloagă et al. (2022), as both stations are geographic located in warm regions. In our case, average surface temperature is 25.2 $\pm$ 5.5 °C in summer, by contrast with 17.4 $\pm$ 6.5 °C in winter. For monthly averange surface relative humidity, values appear slightly higher than those documented by Nomokonova et al. (2019), with 50 $\pm$ 21% in summer and 64 $\pm$ 17% in winter.

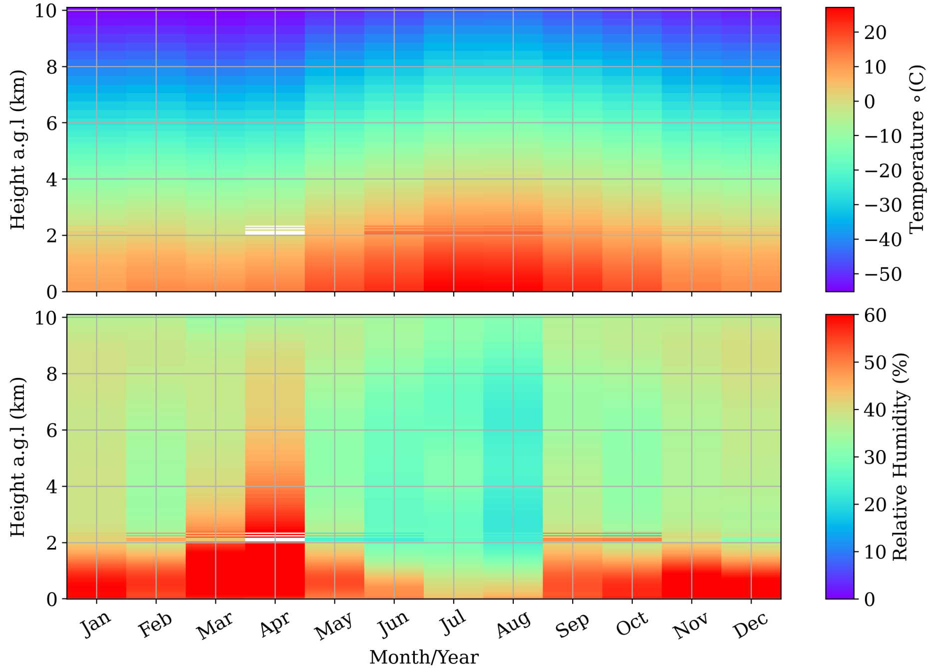

Figure 2. Montlhy evolution of average temperature and humidity profiles obtained with the microwave radiometer.

Following this, hydrometeor formation and consequently, cloud formation will be directly affected by thermodynamic conditions in these regimes. For tracking this changes on clouds, a first approach on monthly hydrometeors occurrence is depicted in Figure 3. Regarding the 6-year CLOUDNET post-processed database, this figure shows the monthly frequency profiles of liquid (top panel) and ice (middle panel) hydrometeors. The lowest panel also displays their corresponding monthly frequency of occurrence, assigning one occurrence per profile if at least one hydrometeor was detected in that profile. In addition, right panels shows the frequency profiles through the entire dataset for each season.

Notably, monthly evolution of ice and liquid hydrometeors keeps aligned with temperature and relative humidity profiles. For both cases, higher frequencies were encountered in the end of cold regimes (November) and during warm regimes (Dec-May). Where we can easily find frequencies ranging 20% for liquid and 15% for ice. Also, a prominent minimum occurred exactly in summer, which can be better visualized in the right panels of Figure 3. These seasonal profiles showed an increase for the liquid case from summer to winter, and for the ice case, an increase from summer to fall, keeps almost the same in winter, and reach a maximum in spring. This feature is strong for the liquid case, where there is no difference between winter and spring. Also relevant frequencies for this case is below 2 km. On the contrary, middle panel of Figure 3 shows that ice particles have a spread distribution on the vertical, overcoming 10 km with slight maxima in 2 km and 8 km.

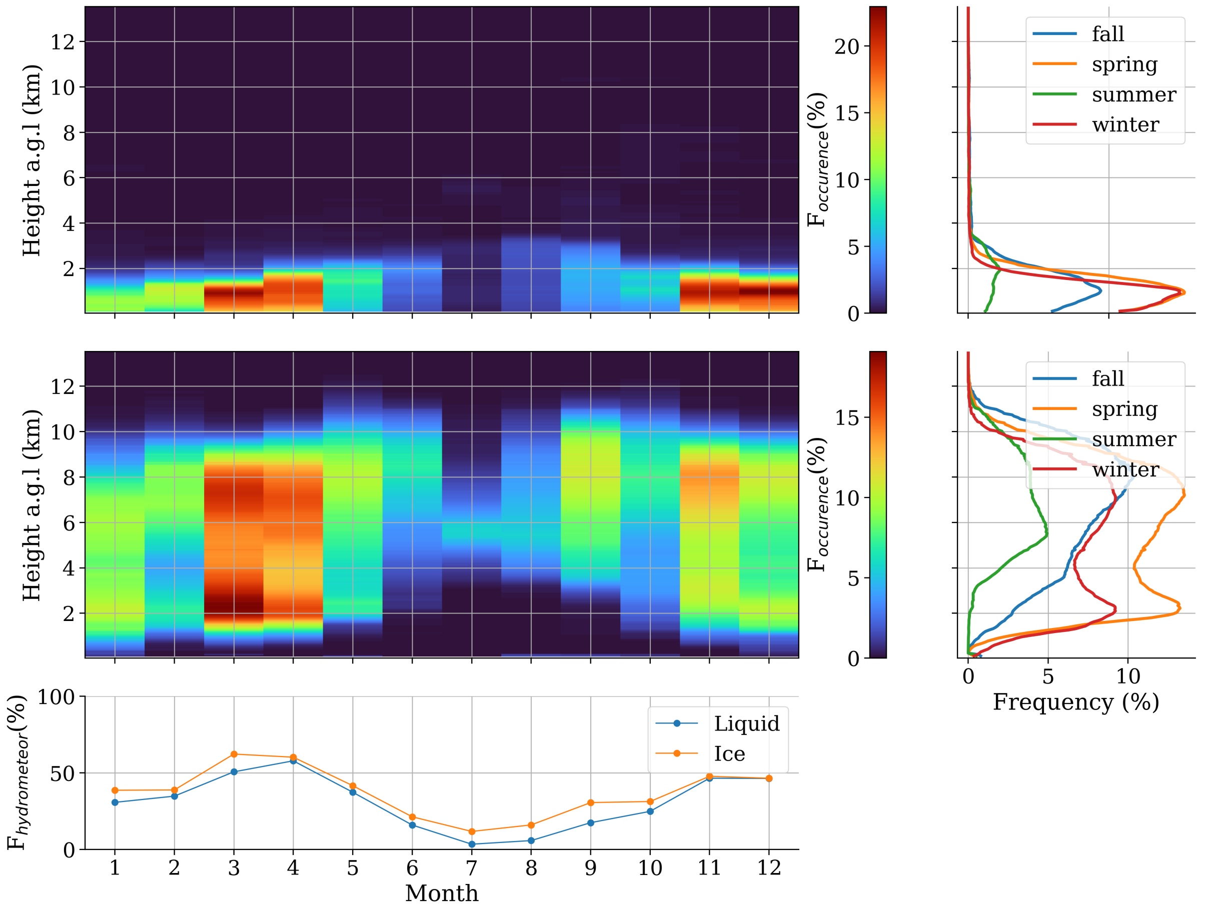

Figure 3. ...

4 Single layer cloud properties evaluation

4.1 Cloud occurrence and microphysics properties

By converting hydrometeors into cloud types using the cloud classification algorithm (refer to section 2.3), monthly frequency of single-layer clouds occurrence was obtained as depicted in Figure 4. Indeed, all subsequent analyses focus on single-layer clouds, as the CLOUDNET target classification product can be highly inaccurate in multilayer cases due to errors in LWP partitioning (Nomokonova et al., 2019). In this scenario, this figure shows a predominant presence of precipitating mixed-phase clouds (Indicated as Pre-Mixed-Phase). Alternatively, almost no precipitation due to liquid clouds was observed, just a slight maximum of 5% in the transition period between fall and winter. Next, ice clouds are the second most present cloud type, followed by mixed-phase clouds and liquid clouds. However, the presence of precipitating mixed-phase clouds are significantly higher than other cloud types, with a substantially stronger seasonal pattern. Its maximum frequency coincides with the cold regimes, in winter and spring, where their frequency reaches 25% and 30%, respectively. Evidently, the minimum occurs in summer (June - August) for all cloud types, when temperature is high and relative humidity is much lower. Also, during summer ice clouds prevailed and no significant liquid precipitation was registered.

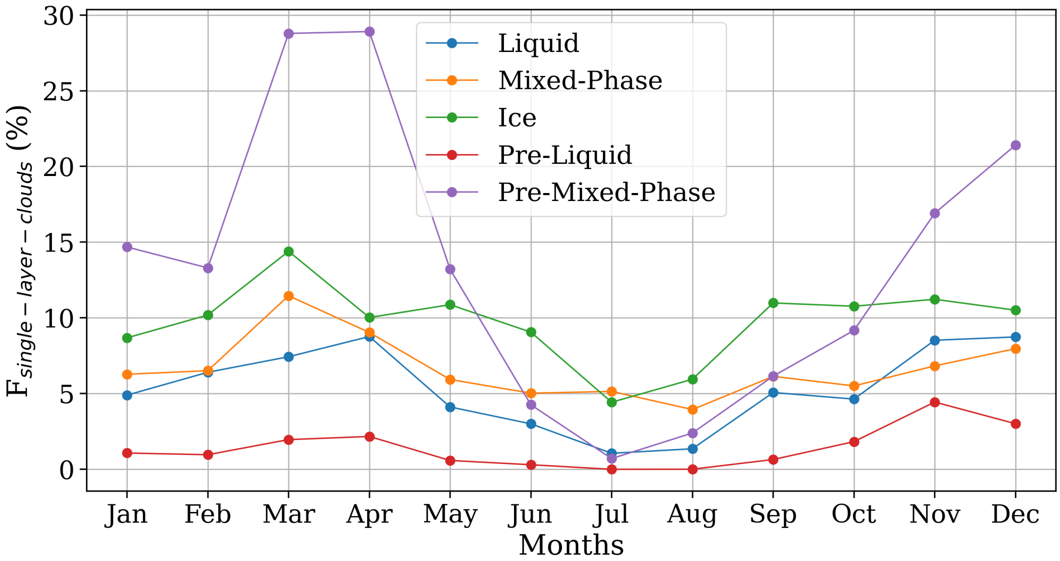

Figure 4. Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

Next, seasonal evolution of cloud thickness (ΔZ), base and top heights are displayed in Figure 5. In relation to cloud thickness, mixed-phase and precipitable mixed phase clouds exhibit the largest and most variable values, followed by ice, precipitating liquid clouds and liquid clouds. A notably broader distribution for precipitable mixed-phase clouds were observed in summer, with a median of 3 [0.6 to 5.1] km. Where values in brackets represents the 25^{th} and 75^{th} percentiles, which demonstrate a greater uncertainty for this variable. Moreover, in this season, these clouds have less than 4% of occurrence (refers to figure 4). Throughout the remainder seasons, median cloud thickness for precipitable mixed-phase clouds are raging from 1 to 1.4 km, close to non-precipitable clouds thickness, ranging from 1.4 to 1.7 km during the entire period. The significant high values Annually, Ice cloud thickness range from 0.7 to 0.9 km, precipitable liquid clouds from 0.27 to 0.8 km, and liquid clouds, from 0.18 to 0.2 km. In general, cloud-thickness distribution is skewed toward large thickness throughout all seasons, and the mode is consistently smaller than the median. Additionally, there is no evident seasonal pattern for this parameter due to its considerable variability.

Subsequently, the middle panel of figure 5 shows that ice clouds have the highest cloud base height, followed by mixed-phase, precipitable mixed phase, and both liquid clouds, precipitable and non-precipitable. In summer, cloud base distribution experiences the most pronounced alterations. For instance, liquid clouds exhibited a wider and higher spread distribution, with a median height of 3 [1.8 to 4.9] km. In contrast, their median ranged from 1.4 - 1.8 km for the remaining period. In relation to mixed-phase clouds, its base height distribution becomes sharper in summer, with a median height of 5.2 [4.6 to 6.0] km, practically the same of fall, at 5.4 [4.0 to 7] km. For the rest of the seasons, their median height is approximately 1 km lower, between 3.7 and 4.0 km. Moreover, precipitable mixed-phase cloud bases are significantly higher in summer, with a median height of 3 [1.8 to 4.9] km, in contrast with the remaining seasons, varying between 1.2 - 1.9 km. Precipitable liquid clouds also presented a higher and broader cloud base in summer, with a median height of 1.7 [1.8 to 4.9] km, about 1 km higher than other seasons, which fluctuates between 0.7 to 1.2 km. For ice clouds, despite exhibiting a higher median cloud base in fall, during summer, an uncommon mode in comparison with other seasons is observed at 5 km - indeed, this is also observed for liquid and mixed-phase clouds. During fall, ice cloud base height exhibits a median value of 8.1 km [6.7 to 9.3] km, also 1 km higher than other seasons, quite well stable in 7.0 km.

Bottom panel of Figure 5 shows a similar seasonal pattern for cloud top height, where the main difference was observed in summer. Particularly, for precipitable mixed-phase clouds, cloud top height has substantially greater variation than cloud base height, with skewed distributions toward large heights. Evidently, this variability reflects in cloud thickness as previously observed. Thus, cloud thickness is directly affected by cloud top measurements in precipitable clouds. Since the CDR reflectivity experienced greater attenuation in this case, cloud top height could be underestimated in some cases, leading to the observed variations. For these clouds, median cloud height, normally encountered between 2.3 - 3.7 km, were found in 6.2 [3.4 to 8.4] km in summer. Similarly, precipitable liquid clouds which have a median top height around 1.4 - 1.7 km, reached at 2.2 [2.0 to 2.6] km in summer. Alternatively, ice and mixed-phase clouds reached a maximum median height in fall of 9.5 [8.3 to 10.4] km and 8.2 [6.3 to 9.6] km, respectively. For the rest of the period, their median top height stayed between 8.5 - 8.7 km and 6.3 - 6.7, respectively. Subsequently, liquid clouds presented larger top height variability in summer, similar to its cloud base height distribution. During this particular season, their median top height is 2.9 [1.8 to 4.9] km, roughly 1 km higher than other seasons, which are varying from 1.4 - 1.8 km.

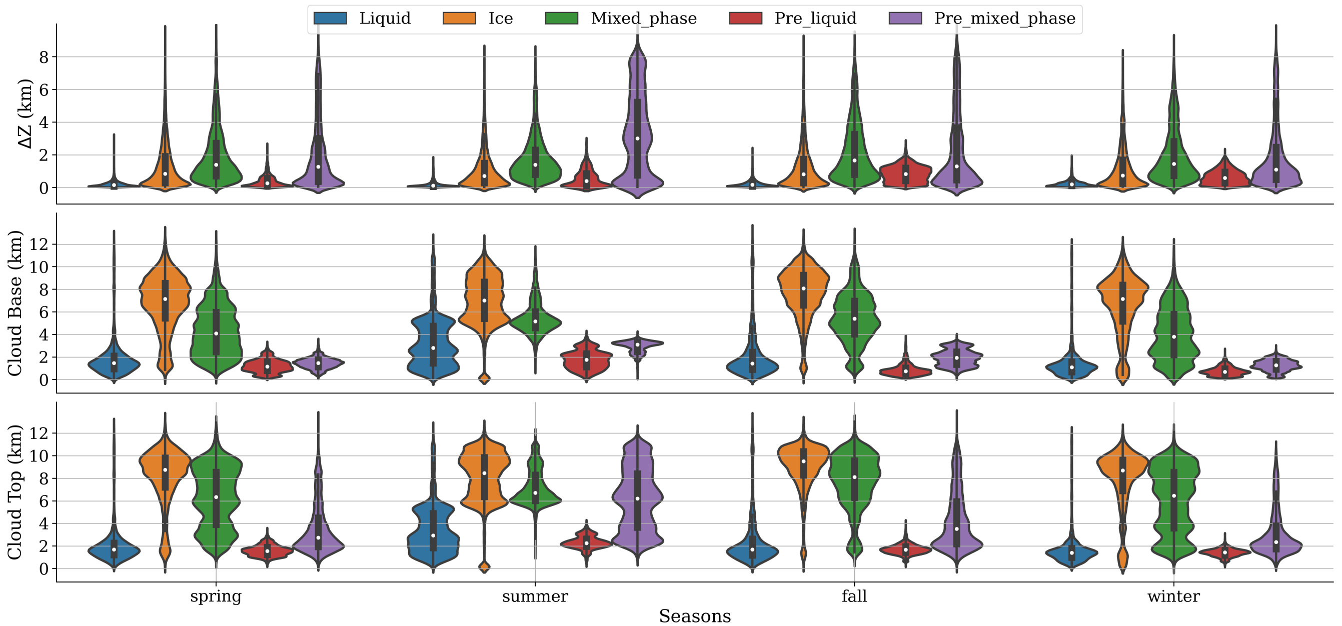

Figure 5. Seasonal evaluation of cloud thickness, base and top. The legend above indicates the color correspondent to each cloud type. This figure shows a dark gray box-plot with a white median inside an estimated density distribution. This distribution is displayed in relation to the middle of the box-plot in the right and the left direction.

In general, cloud base and top height is higher in summer for almost all cloud types, except for ice and mixed-phase clouds, which reach their peak during fall. Despite that these properties showed a seasonality in terms of median values, the interquartile range showed strong variability, especially for ice and mixed-phase clouds. Specifically, precipitable mixed-phase clouds exhibited heightened variability in cloud top height, notably less precise compared to their base height. This suggests a possible underestimation of cloud tops due to radar reflectivity attenuation by rain droplets. Same behavior would be observed in precipitable liquid clouds, but they are normally thinner and easier to be totally penetrated by the radar signal. Also, these clouds do not have enough occurrence to strengthen the statistics of their macrophysical properties, as can be seen in Figure 4.

4.2 Liquid water content and liquid water path

Monthly and seasonal evolution of liquid water content (LWC) and liquid water path (LWP) for liquid clouds are displayed in Figure 6. Left panel shows the mean seasonal profiles of LWC, and the right panel shows the monthly mean profiles of LWC and monthly mean values of LWP (right-bottom panel). Since the monthly median of LWC is usually zero for all cloud types, suggesting a dry atmosphere with a typical water shortage, we decided to analyze average values. First, mean seasonal profiles showed peaks of LWC about 0.07 $g\ m^{-3}$ at 1 km, 0.1 $g\ m^{-3}$ at 1.1 km, and 0.14 $g\ m^{-3}$ at 1.7 km during fall, winter and spring, respectively. In spring, the average value at 4 km was strongly affected by the presence of outliers, identified in May by the monthly distribution on the right panel (note the dark red tone), creating an abnormal peak at this level. Despite that summer showed a sharp peak of about 0.16 g m $^{-3}$ at 200 m, above this level, this season has lower average LWC in comparison with other seasons. In addition, the monthly evaluation of LWC, displayed in the right panel, points out its tremendous variability, which passes through several orders of magnitude. This panel shows that relevant LWC are below 2 km, in addition its average decreases two orders of magnitude in July and August (during summer) but are well vertically distributed, reaching 5 - 6 km. This spatial arrangement is in agreement with the cloud base and top shown in Figure 5. Also, the presence of small values occur much more frequently than large values, leading to several outliers which are affecting the mean LWP, as can be seen in the bottom-right panel of figure 6. The mean LWP is denoted by the black star, and the median by the box-plot red line. A large discrepancy between these two statistical parameters was observed, except in summer where values are all small and there are relatively low cloud liquid profiles, as indicated by the blue lines. In general, average LWP stays below 100 g m^{-2} for liquid clouds, with 78% of frequency of occurrence and exhibit a clear seasonal pattern, with minimum in summer and maximum in winter-spring. On the other hand, the median does not exhibit a significant seasonal pattern.

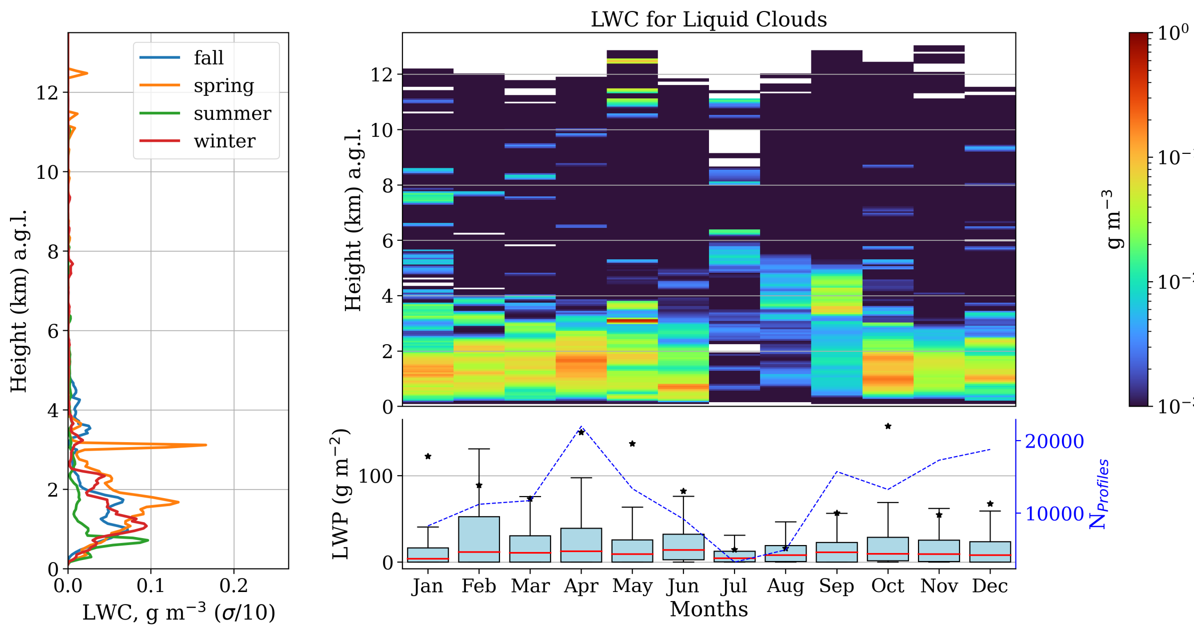

Figure 6. Seasonal and monthly average of liquid water content (LWC) and liquid water path (LWP) for liquid clouds. Left panel shows the seasonal mean profiles of LWC. Right panel shows the monthly mean profiles of LWC and monthly box-plot of LWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Similar seasonality was observed for precipitable liquid clouds, depicted in Figure 7. Nevertheless, LWC are higher due to the presence of drizzle, and peaks are more sharply defined around 1 km during fall, winter and spring. Also, winter achieved higher values, twice as large as in fall and spring (0.2 g m^{-3}), reaching 0.7 g m^{-3}. In summer, values are relatively smaller, with peak of 0.1 g m^{-3} around 2 km. Right panel shows that monthly values still fluctuate through several orders of magnitude, and larger values are not well vertically distributed in summer as observed for the only liquid case. They still kept within thin layers, although they reached at slight higher levels. This is in agreement with vertical position of cloud base and top height (see Figura 5), which does not showed a significant vertical variability in summer. In terms of LWP, no seasonality was observed due to the presence of rain droplets, which highly increased its water content in all months. Note that LWP were not separated between cloud and rain, it is the integrated value reported by the microwave radiometer. As a result, slight discrepancy between the mean and the median was observed, and its values are roughly 10 - 20 times higher in comparison with the liquid case, with 75% of data below 250 g m^{-2}. Also, October presented a significant variation of LWP in relation to other months, with more than 1000 g m^{-2} inside the 75^{th} percentile. Thus, this month is marked by stronger rains from this particular cloud.

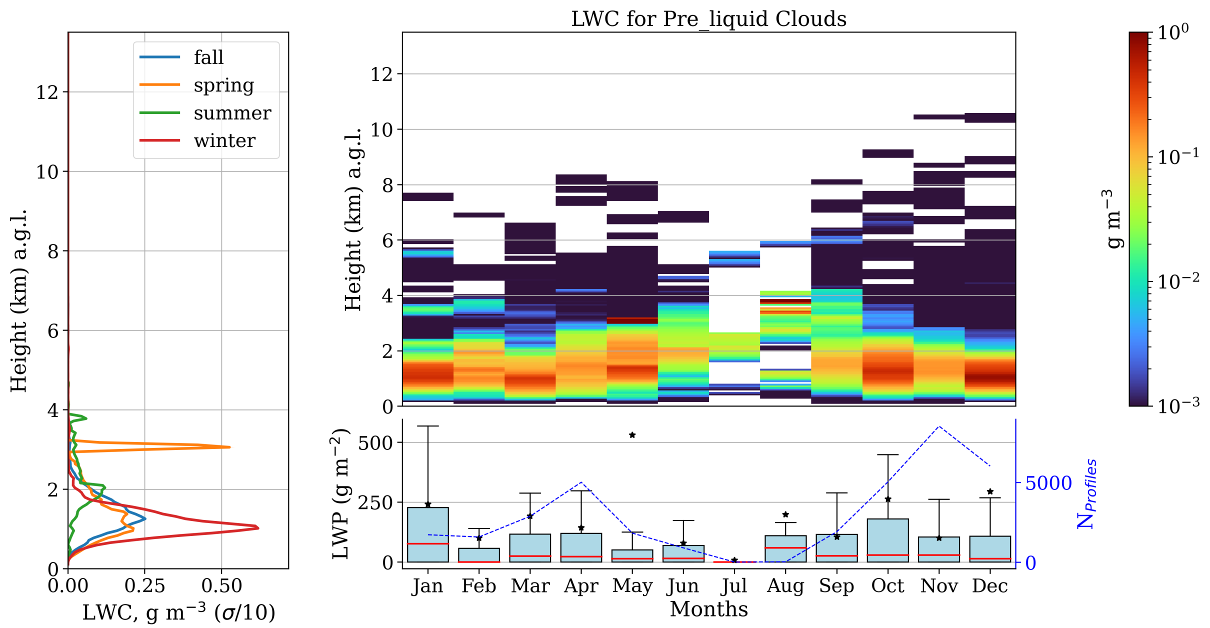

Figure 7. Same as Figure 6, but for precipitable liquid clouds.

Mixed-phase clouds exhibited marginally greater occurrence than liquid clouds (Figure 4). Typically, they co-occur and can be comparable in terms of water content. Seasonal average profiles of LWC for mixed-phase clouds in the left panel of Figure 8 shows a similar vertical distribution of LWC between summer and fall (warm regimes), and between winter and spring (cold regimes). Below 4 km, cold regimes have higher average LWC, and similar values with warm regimes above it. In general, these profiles have just smaller differences, and average values are always less than 0.05 g m^{-3}. This shows that liquid clouds, besides being thinner, have more water than mixed-phase clouds, suggesting that mixed-phase clouds are mainly composed by ice hydrometeors. Next, monthly mean time evolution on the right panel shows a spreader vertical distribution of LWC, which becomes thinner and higher in summer. This suggests that liquid layers inside mixed phase clouds become thinner in summer, and the ice content is still high enough to keep its thickness around the seasons, as observed in figure 5. Additionally, the right-bottom panel of Figure 8 shows greater occurrence of large LWP values at the end of winter and during spring. Overall, 84% of the analyzed mixed-phase cloud profiles have LWP below 100 g m^{-2}, emphasizing the point these clouds have less water then liquid clouds, which has 78% as already discussed.

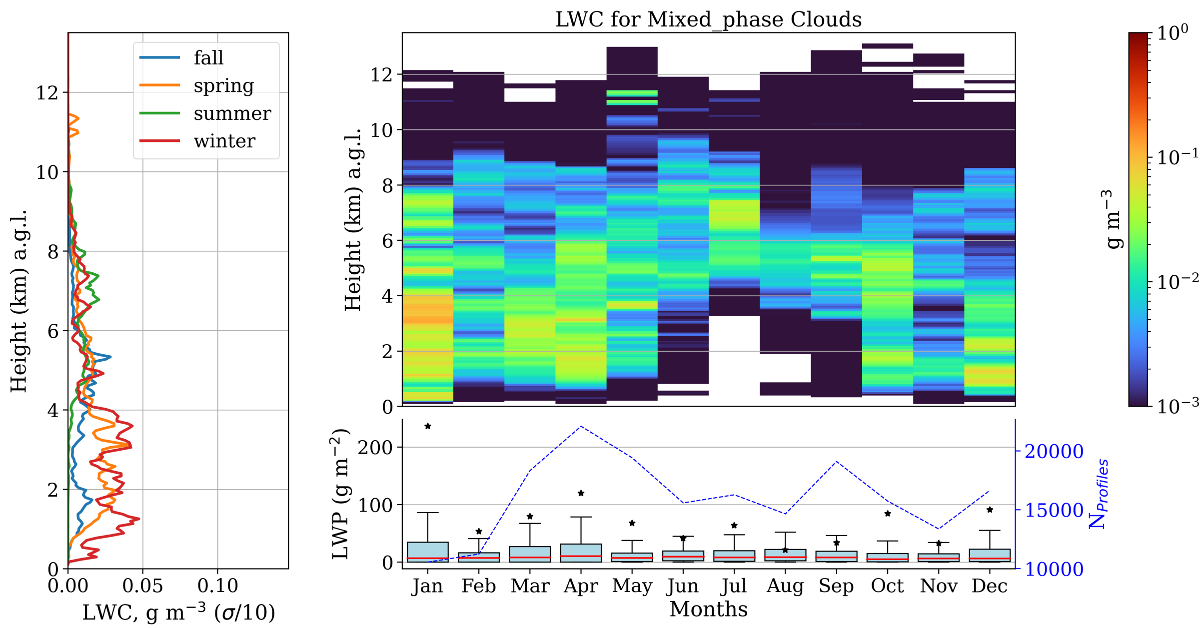

Figure 8. Same as Figure 6, but for mixed-phase clouds.

Left panel of Figure 9 reveals that precipitable mixed-phase clouds present similar average profiles up to 4 km in fall, winter and spring. During summer, average LWC within this layer are more than twice smaller in comparison with other seasons. On average, these clouds still have less water content than precipitable liquid clouds below 2 km. This evidence, combined with the observation that liquid layers are notably narrower than mixed-phase layers, suggests that liquid hydrometeors represent a mere fraction of mixed-phase clouds. This observation has more weight here since precipitable mixed-phase clouds have much more frequency of occurrence. In addition, the blue axis in the bottom-right panel of Figure 9 shows a relevant higher number of cloud profiles, with a sudden decrease in summer. Monthly evolution of LWC clearly indicates a concentration of LWC below 4 km, except in summer, when these clouds have a higher cloud base. In general, the observed mean values are higher than only mixed-phase clouds due to the elevated presence of drizzle. In this case, LWC has a narrower vertical distribution and is closer to the ground. Therefore, they will also have much higher LWP than mixed-phase clouds, as depicted in the right-bottom panel. In terms of its monthly median, no clear seasonal pattern is identified. However, as in the precipitable liquid case, a strong variability of LWP was found in October. This time, we can assume with more statistical confidence that rain events in October are more intense. Although precipitation is more frequent during spring. Moreover, precipitable mixed-phase clouds exhibit 70% of LWP below 250 g m^{-2}; in other words, they can attain higher values than precipitable liquid clouds.

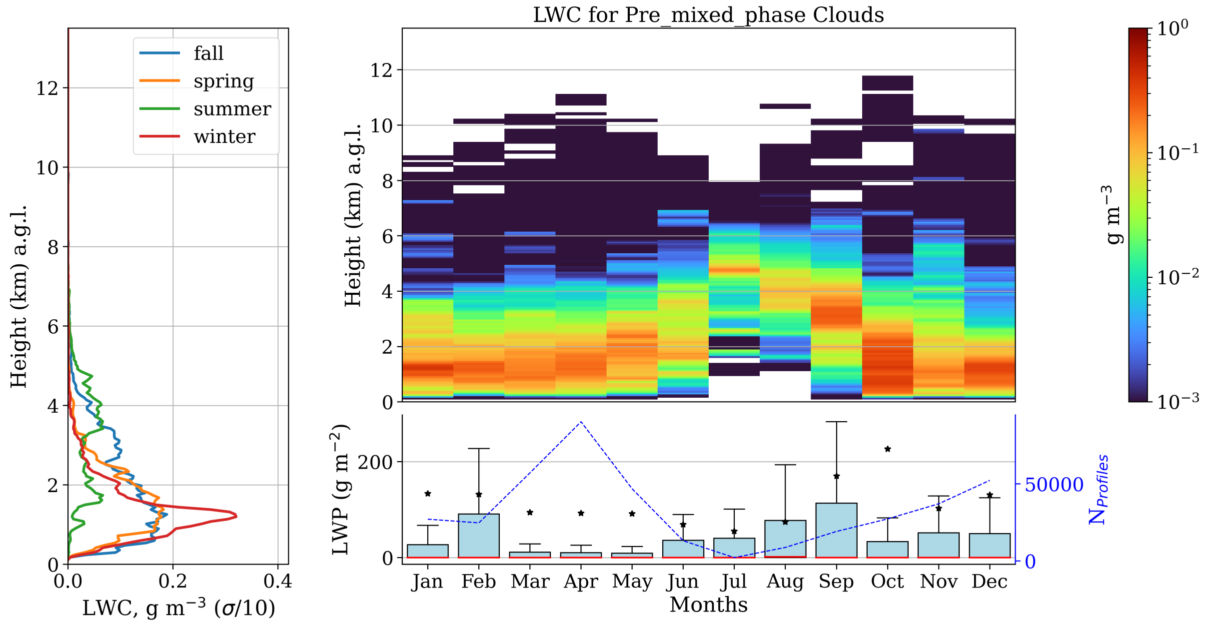

Figure 9. Same as Figure 6, but for precipitable mixed-phase clouds.

Overall, a notable reduction on LWC was observed in summer. Also, values were found at high levels in the atmosphere in this season, in accordance with the cloud base height pattern. These differences were more readily observed by mixed-phase and precipitable mixed-phase clouds. On the other hand, no seasonal pattern of LWP was clearly identified for all clouds, owing to the high month variability of this variable. To conclude, no clear seasonal pattern was identified for clouds in relation to monthly median LWP. However, liquid clouds have greater LWP than mixed-phase clouds. Once mixed-phase clouds are considerably thicker than liquids, the majority of their composition comprises ice hydrometeors, which will be confirmed in the sext section. Moreover, for precipitation cases, the liquid water content increases due the presence of drizzle, and LWP can reach values approximately 10 times greater. In addition, precipitable mixed-phase clouds can achieve higher values of LWP than precipitable liquid clouds.

4.3 Ice water content and ice water path

Figure 10 displays the average IWC content profiles for each month and seasons, and monthly box-plots of IWP. An evaluation of seasonal profiles on the left panel of this figure denoted a markable presence of IWC at 2 km in winter, with a peak of 0.2 g m^{-3}, while at the same level, no IWC was observed during other seasons. Therefore, despite that Figure 5 shows almost no ice cloud base at this level during winter, the presence of just a few clouds with high ice water content are affecting its average. In fact, monthly profiles in the right-top panel reveal that this ice cloud occurs in January. This can be seen by the red tone centered exactly at 2 km in this month. During spring, the peak is around 0.12 g m^{-3} at 4.5 km, and above this level its average profile coincides with the winter profile. A similar feature is observed for warm regimes, average profiles of fall and summer coincide above 7 km. Below this level, fall has a peak of 0.2 g m^{-3} for about 5 km and summer has a peak of 0.1 g m^{-3} around 6 km. In general, ice cloud base heights are usually above 6 km (refers to Figure 5). Hence, relevant values of IWC for these clouds are within 6 and 10 km, in addition, there is no IWC below 2 km, except in January. Concerning the IWP, the median are relatively smaller than the mean, showing a right-skewed distribution with outliers, which are affecting their monthly mean. In terms of medians, no clear seasonal pattern was found. However, larger variability was observed during spring and beginning of fall. In fall (Sep-Nov), ice cloud bases are higher, and average IWP also experienced a maximum of about 100 g m^{-2} in September. Regarding the entire dataset, 90% of the IWP are smaller than 100 g m^{-2}.

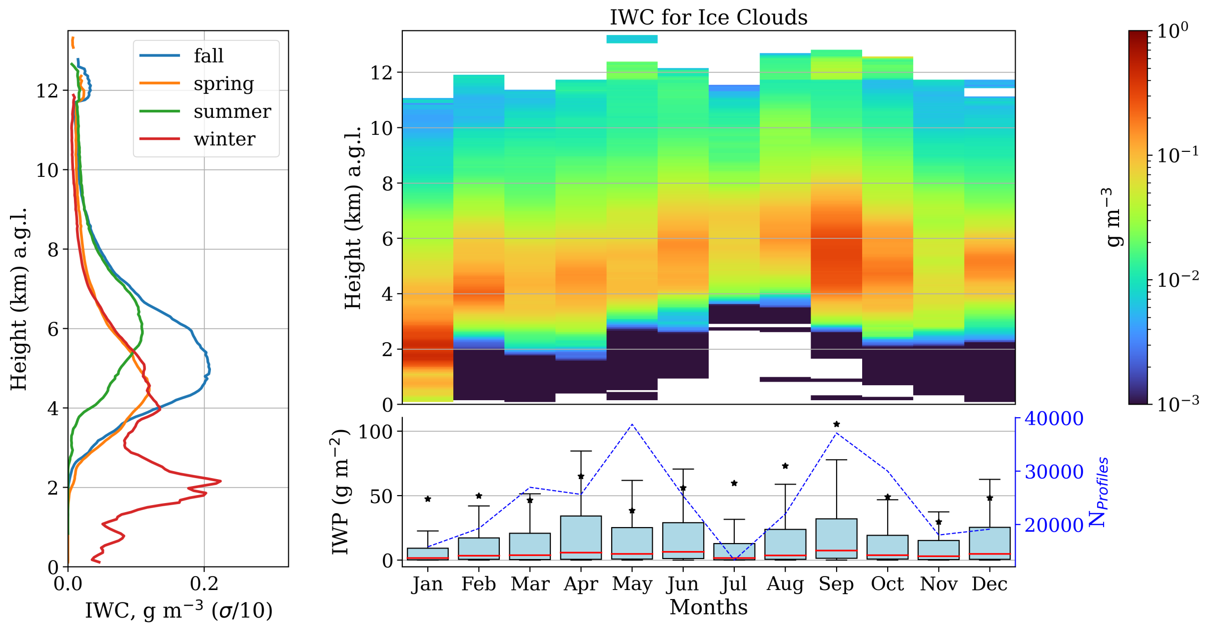

Figure 10. Seasonal and monthly average of ice water content (LWC) and ice water path (LWP) for ice clouds. Left panel shows the seasonal mean profiles of IWC. Right panel shows the monthly mean profiles of IWC and monthly box-plot of IWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Mixed-phase clouds exhibited a similar mean profile of IWC during summer and fall as can be seen in the left panel of Figure 11. The only significant difference was during the summer, when mixed-phase clouds do not have a peak around 2 km as observed for ice clouds. Although seasonal profiles suggest mixed-phase clouds have almost the same amount of IWC of ice clouds, it is worth noting that cloud thickness for ice clouds is smaller than for mixed phase clouds. It was observed that mixed phase clouds are almost 1 km thicker than ice clouds. Thus, IWC will be greater, which is the integrated IWC, will be greater as can be seen by the box-plots in the right-bottom panel. This panel shows that IWC easily surpasses 50 g m^{-2}, which was not common for ice clouds. In addition, there is less discrepancy between monthly mean and median than for the ice case. Also, 74% of all IWP for mixed-phase clouds are below 100 g m^{-2}, showing that these clouds have more ice content than ice clouds.

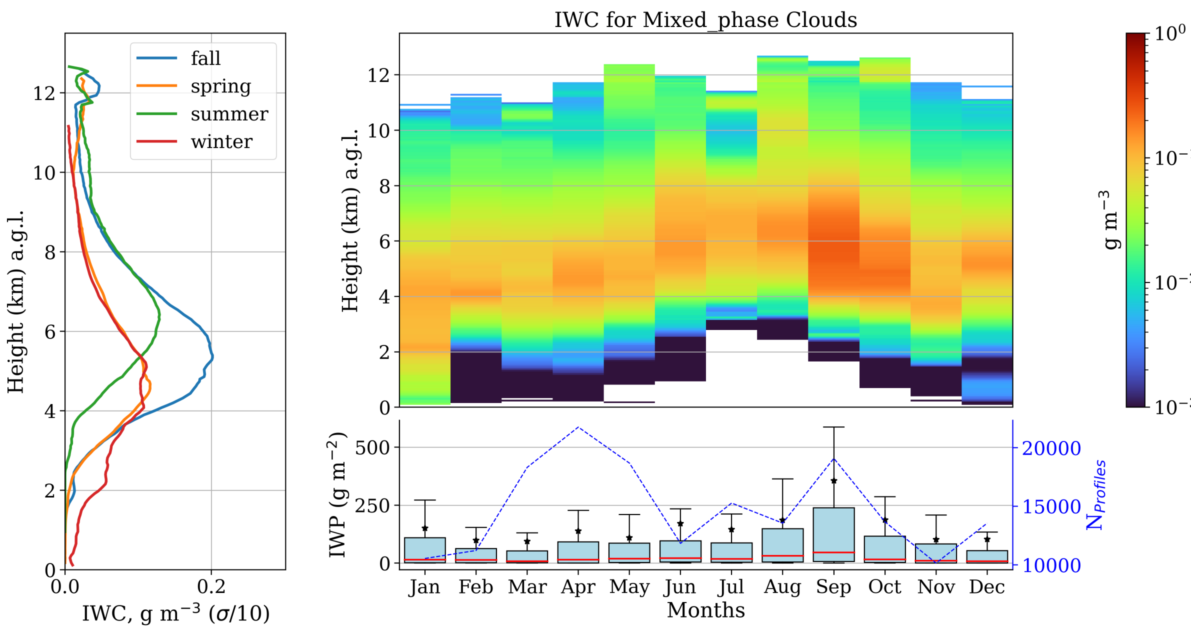

Figure 11. Same as Figure 10, but for mixed-phase clouds.

For precipitable mixed-phase clouds, winter and spring still have a similar mean profile of IWC (see Figure 12), with peaks around 0.2 g m^{-2} at 4 km. Fall still with a peak around 5 km as observed for the ice and non-precipitable mixed-phase clouds, with a just slightly higher value. The most considerable changes happened in summer, where the average profile reached values higher than the other seasons around 6 km, and kept higher up to 12 km. Within this layer, the right panel indicates larger values during summer (Jun-Aug), in red tone. As a result, IWP in this season is significantly higher in comparison with the rest of the period (see right-bottom panel of Figure 12). Although precipitable clouds have much less frequency of occurrence in summer, this frequency is just 4% smaller than mixed-phase clouds. Also, precipitable mixed-phase clouds exhibited a larger cloud thickness, almost 2 km greater than just mixed-phase clouds in this season. By comparing cloud thickness based on similar cloud frequencies, we can infer that precipitable mixed-phase clouds exhibit significantly greater thickness and substantially larger IWP, particularly during the summer and September. Thus, despite the fact these clouds have extremely low frequency in summer, they demonstrate greater vertical development and contain significantly more ice compared to other periods

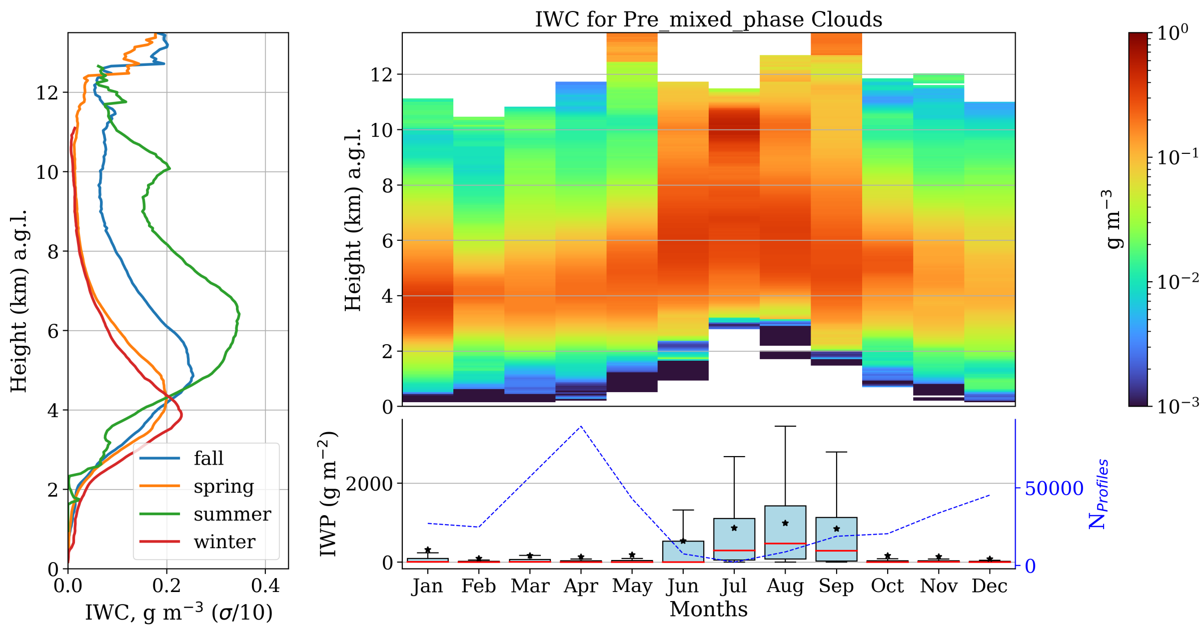

Figure 12. Same as Figure 10, but for precipitable mixed-phase clouds.

To summarize, mixed-phase clouds presented higher IWC than ice clouds. The reason can be attributed to the large thickness of the mixed-phase case compared to the only ice case, which is approximately twice as thicker. Thus, less than half of mixed-phase clouds are made of liquid hydrometers, and the rest are ice hydrometeors. Also, precipitable mixed-phase clouds showed even higher contents of ice, especially in summer. Beside their poor frequency of occurrence in summer, these clouds boast significantly greater thickness compared to mixed-phase clouds, carrying substantially more ice.

4.4 Droplet effective radius

For liquid clouds, droplet effective radius (r_e) were also analyzed by season and monthly. The following analysis was made considering only the median of r_e and not average values. Left panel of Figure 13 shows the seasonal vertical distribution of the median droplet effective radius where the shaded area represents the 25^{th} and 75^{th} percentiles. Right-top panel shows the monthly vertical distribution of the median, and the right-bottom panel shows the median and an estimation of the probability density function (p.d.f) of r_e by month using violin plots. These estimations were made through kernel density method **?** and the p.d.f are shown with its reflection with respect to the middle point in the x-axis, with a kind of violin shape. Additionally, this panel has a second y axis with the number of profiles at each month to compute statistics. First, no seasonal difference was observed for profiles on the left panel of Figure 13. Second, monthly profiles on the right-top panel show that large droplets are found above 4 km, in yellow and red tones. However, monthly examination r_e distribution in the bottom panel demonstrates that the medians are about to 5 μm, corresponding to the soft blue tone below 3 km of height, consistent with the LWC distribution in Figure 6.

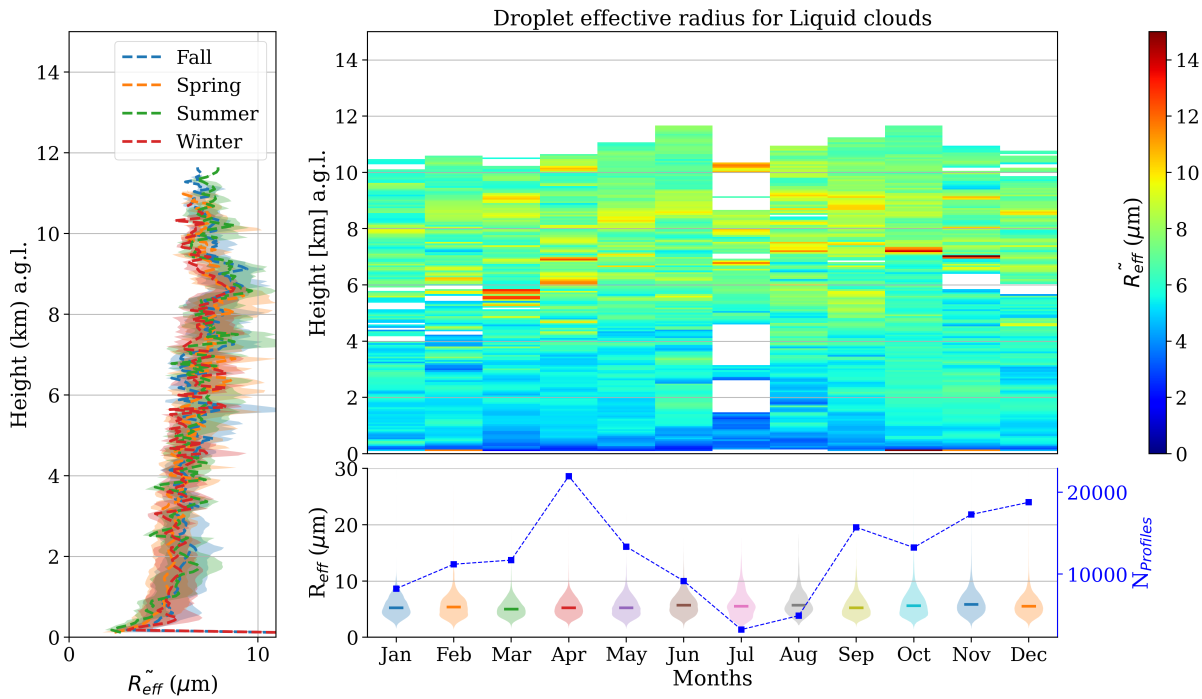

Figure 13. Seasonal and monthly statistics on droplet effective radius (r_e) distribution for liquid clouds. Left panel shows the seasonal median profiles of droplet effective radius, where shaded area denote the 25^{th} and 75^{th} percentile. Top-right panel shows the monthly profiles of the median r_e, while the bottom-panel displays the size distribution of r_e by months.

Next, precipitation liquid clouds have a slight difference on seasonal profiles below 2 km, as can be seen on the left panel of Figure 14. Below this level, spring has the lowest values, followed by winter, fall and summer. However, in this case, there are almost no profiles with precipitable liquid clouds in summer. This can be seen by the blue line in the right-bottom panel. Thus it is challenging to evaluate seasonal differences on liquid droplets for these clouds which have almost zero frequency of occurrence in summer. On the other hand, it is clear that their liquid droplets are greater than for liquid clouds. This can be associated with the presence of drizzle, as discussed in terms of LWC. In this case, the median droplet effective radius ranges around 8 and 9 μm, except in summer, where droplets are greater than 15 μm.

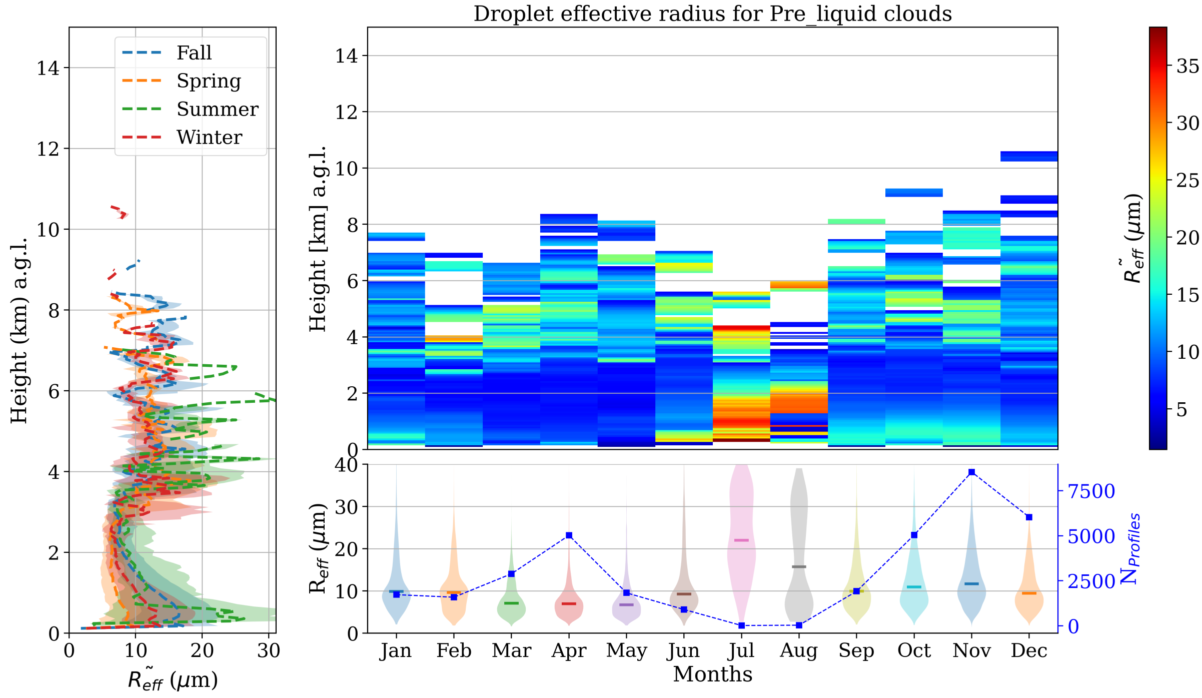

Figure 14. Same as figure 13, but for precipitable liquid clouds.

Concerning mixed-phase clouds (Figure 15), no seasonal differences were observed. Nonetheless, Left panel of Figure 15 shows that droplets of mixed-phase clouds are systematically larger than within liquid clouds, roughly twice the size. In addition, larger cloud droplets are found between 3 and 5 km of height. This is also observed by the monthly median profiles in the right-top panel, by the yellow and red tones. Same panel also indicates that greater droplets sizes are found in summer, despite the fact there are small amounts of LWP in this season as observed in Figure 8. Even though an evaluation of droplet size distribution on the right-bottom panel of Figure 15 shows an stable median around 10 μm through the months, droplets can vary between 13 - 22 μm in the neighbourhood of 4.5 km of height in summer. However, values are typically much smaller than this, as can be seen by the monthly p.d.f in the right-bottom panel. They show a mode below the median during the entire year, revealing that the most frequent droplet sizes are even smaller than 15 μm.

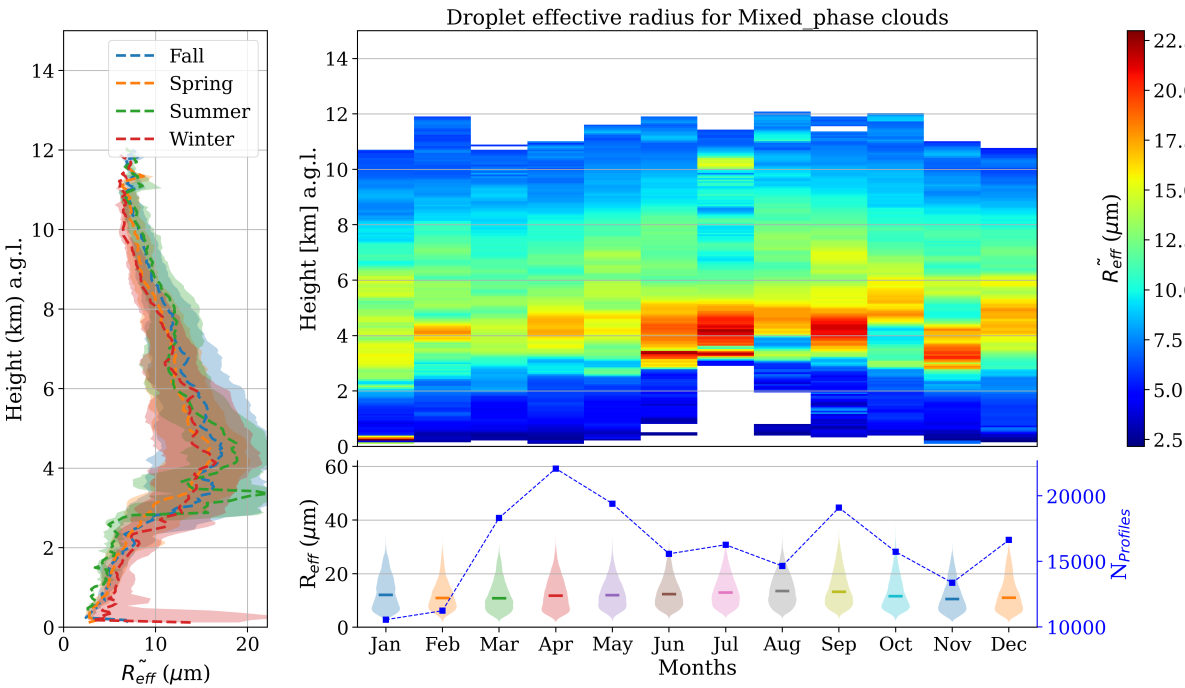

Figure 15. Same as figure 13, but for mixed-phase clouds.

Differently, precipitable mixed-phase clouds do not exhibit vertical variation of droplet size. Their median values were kept almost the same along the vertical by each median profile of Figure 16. Also, summer showed a slightly higher median profile of r_e than other seasons, roughly 10 μm greater than winter, between 3 and 6 km. Differences below this layer are quite challenging to analyze since these clouds have a cloud base about 3 km in summer, as already observed in Figure 5. Thus, besides these clouds having almost no occurrence in summer, there is almost no cloud base below this level. Therefore, a seasonal comparison of r_e should be done above this level, which still has larger r_e, but differences are less discrepant. An evaluation of the monthly distributions on the right-bottom panel also attribute less weight to these low clouds during summer. As a consequence, a difference around 5 μm was also observed between winter and summer medians. In summer, the median is varying around 20 μ, and during the winter, the median is about 15 μm. In addition, during summer it is possible to observe that the position of the mode, which is located below the median for the rest of the period, starts to move until stays above the median in August.

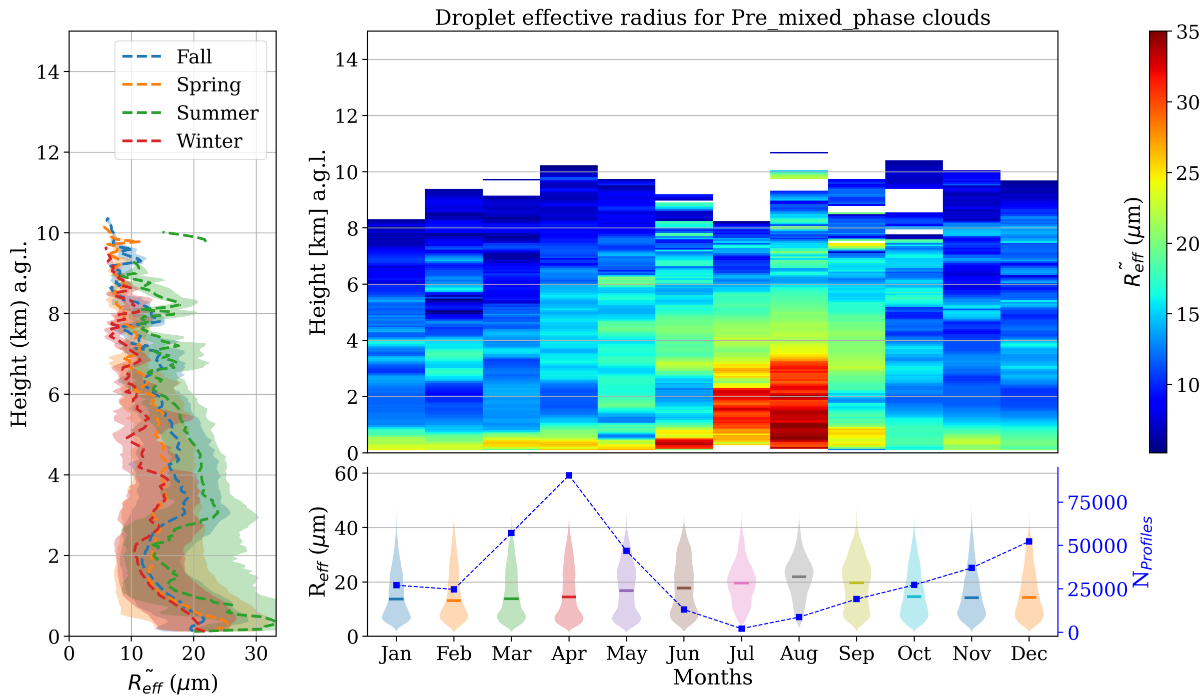

Figure 16. Same as figure 13, but for precipitable mixed-phase clouds.

In general, no strong seasonal differences were observed for liquid and precipitable liquid clouds. Liquid clouds have r_e varying around 5 μm, while precipitable clouds achieved higher values due to the presence of drizzle droplets, varying between 10 μm. Mixed-phase clouds showed consistently higher median r_e compared to liquid clouds, where larger values occurred around 3 and 5 km of height, especially during summer. Since just a few fraction of these clouds have larger r_e, and statistics with all monthly data showed only small variation of the month medians, which stayed around 15 μm. In relation to precipitable mixed-phase clouds, larger differences observed in summer occurred due to the presence of a few low clouds, which were higher r_e. However, these clouds do not have enough occurrence to compare with the other months, which have considerably strong statistics.

4.5 Ice effective radius

Ice effective radius for ice clouds are depicted in Figure 17, and is organized as the previous droplet sizes analysis. Left panel shows a slight seasonal difference above 7 km, where summer assumed larger median r_e. Also, larger values of ice r_e is observed in summer by the yellow tone bellow 4 km in the right-top panel, indicating a radius about 60 μm. However, this clouds have a median cloud base around 7 km in summer, in addition to a 25^{th} percentile of 5 km. This shows that there is a negligible occurrence of ice clouds bellow 4 km. Therefore, r_e for these clouds are typically ranging between 30 and 50 μm, where lower values are more frequently. This can be also seen in the monthly p.d.fs on the right-bottom panel. Throughout the observation period, the median remained close to 30 μm, and increased in summer due to the small presence of low ice clouds.

During this season, median r_e are close to 50 μm. It is worth to note the importance of analysing the median instead of the mean in these cases. Clearly, outliers close to surface, which does not represent a real situation for ice clouds will affect the average in May, August, September and October. The monthly distribution of r_e also shows the presence of a mode bellow the median for almost all season, except for April and summer. As a result, common r_e can be even smaller than 30 μm

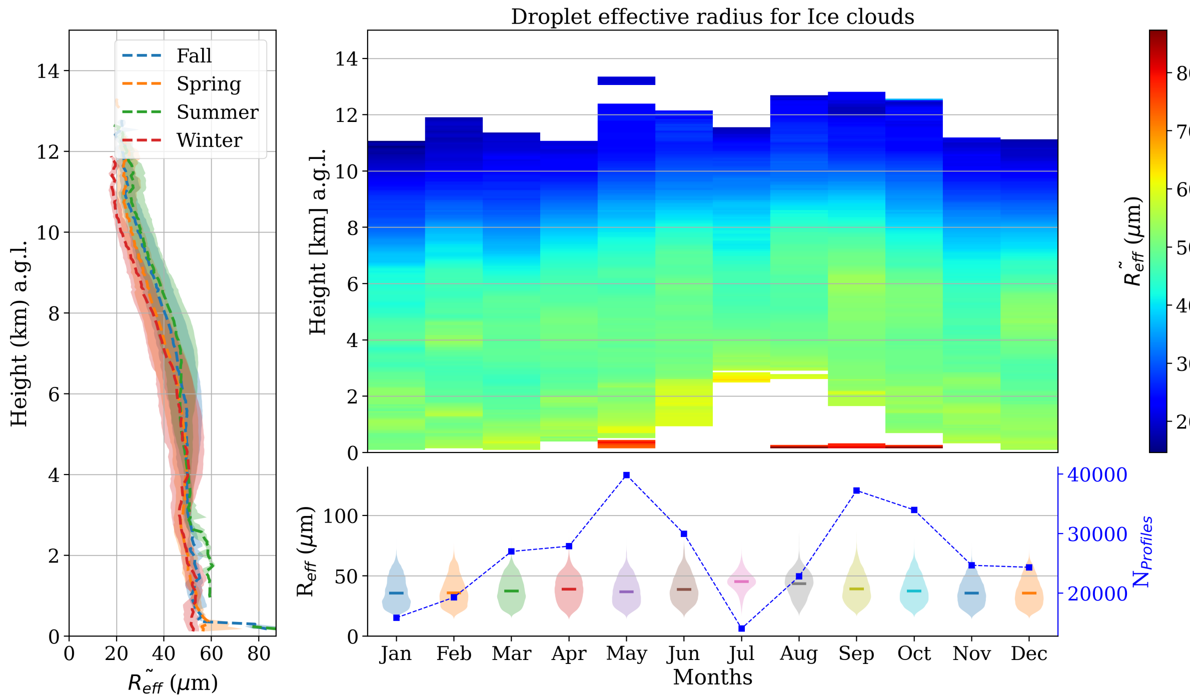

Figure 17. Seasonal and monthly statistics on ice effective radius (r_e) distribution for ice clouds. Left panel shows the seasonal median profiles of ice effective radius, where shaded areas denote the 25^{th} and 75^{th} percentile. Top-right panel shows the monthly profiles of the median r_e, while the bottom-panel displays the size distribution of r_e by months.

For mixed-phase clouds, the left panel of Figure 18 also shows a slight difference between profiles above 6 km. In summer and fall, median values are slightly larger than other seasons. The reason can be associated with the vertical location of clouds at each season, as cloud bases in summer and fall are around 5 km, and one kilometer lower in winter and spring. Hence, high mixed-phase clouds in summer and fall are exhibiting large r_e. Seasonal differences below 3 km are quite challenging to evaluate due to the negligible presence of these clouds. As a reference, the 25^{th} percentile of cloud base height in summer is 4.6 km, denoting a small presence of clouds below this level. However, regarding the entire dataset, the distributions in the right-bottom panel presented slightly larger values in summer and fall, reaching at 50 μm in July, August and September. In general, median r_e are just a little smaller, with high variability. Different from the previous case, the median and the mode almost coincide for all the months.

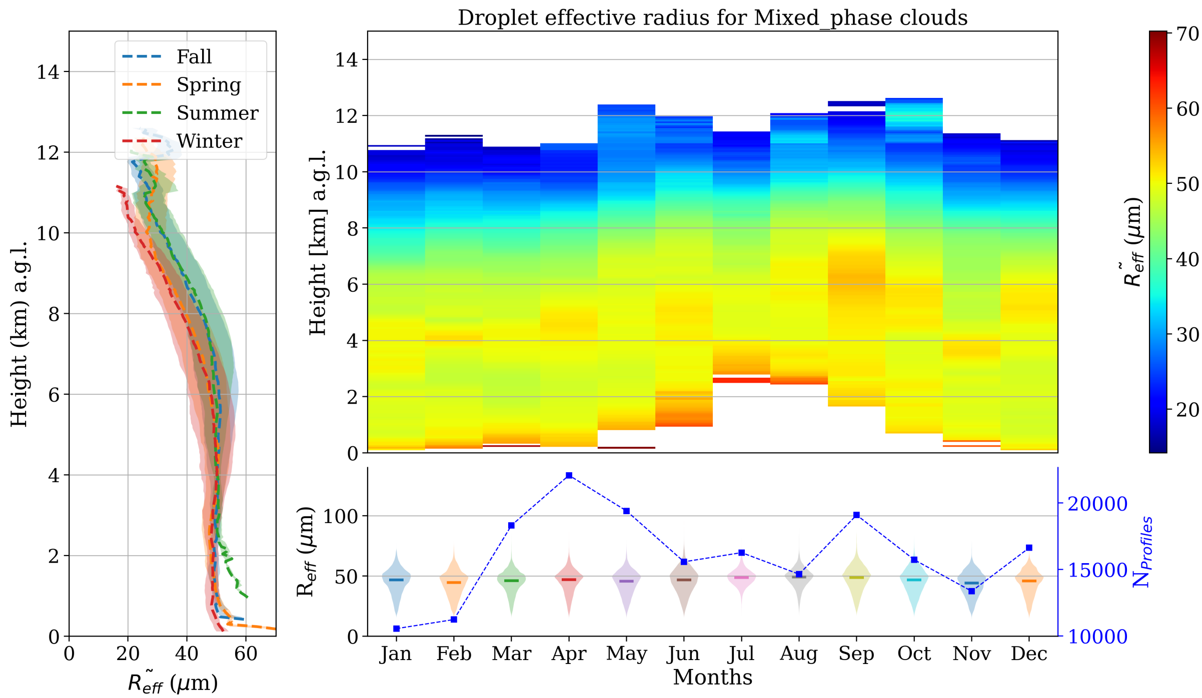

Figure 18. Same as figure 17, but for mixed-phase clouds.

Lastly, precipitable mixed-phase clouds experienced the greater seasonal difference between median profiles. Above five kilometers, summer has the largest median values, followed by fall and then winter and spring with the same values. Differences above 7 km cannot be analyzed since cloud tops are frequently not observed at this level, with the 75 th percentile at 6 and 4.5 km in fall and spring, respectively. Probably, the differences observed above 12 km are due to a bad classification of cloud layers. Same reason can be causing values about 70 μm at 10 km observed in the right-top panel of Figure 18, in July. In addition, this month has a poor number of profiles, denoting just minor cases that can be contributing to these high values. In general, the statistical median of the monthly p.d.fs is stable in 50 μm, with a narrower distribution.

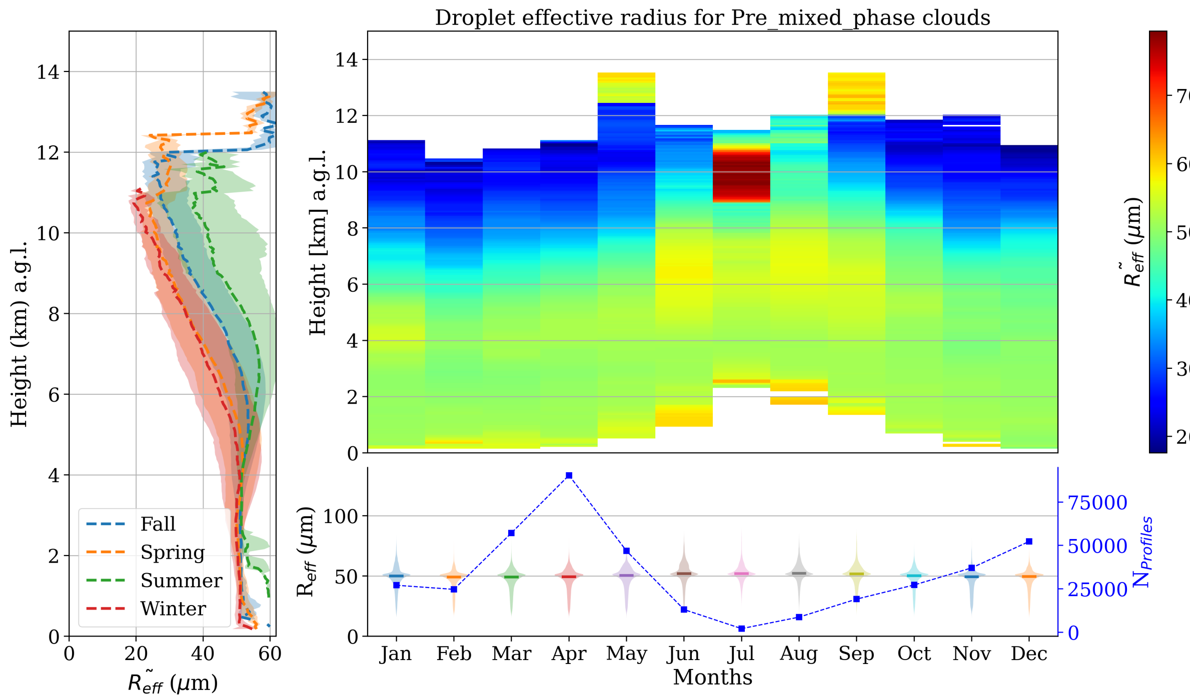

Figure 19. Same as figure 17, but for precipitable mixed-phase clouds.

In summary, ice clouds have median r_e ranging between 30 and 50 μm, where values can be larger at more elevated levels. The second is commonly found during summer, and the first during the rest of the period. For mixed-phase and precipitable mixed-phase clouds, the median r_e are about 50 μm. In fact, mixed-phase clouds have slightly smaller values than 50 μm excluding summer, and a broader distribution. However, values are quite similar with the precipitable case.

5 Summary and conclusions

The study of cloud coverage and its physical properties are the first step in understanding cloud radiative effects. In order to retrieve these variables, CLOUDNET post-processed data of FMCW 94 GHz cloud Doppler radar, HATPRO radiometer and CHM15k ceilometer were used. Additionally, complementary analysis of thermodynamic conditions was performed with the HATPRO radiometer vertical profiles. Temperature and relative humidity profiles tracked cold (fall-winter) and warm regimes (spring-summer). The assortment algorithm was applied using the CLOUDNET target classification product to classify different single layer cloud types. Inter-annual observation on cloud occurrence revealed a seasonal trend, where mixed phase clouds are the prevailing species, followed by ice clouds, especially during spring and winter. In these seasons, precipitating mixed phase clouds can reach 30%XXX and 25%XXX of frequency, respectively, and ice clouds 15%XXX and 12%XXX. Other cloud types have no strong seasonal pattern, and precipitating liquid clouds are absent throughout the year. In addition, the total single layer cloud frequency in summer is less than 8%. Mixed-phase clouds exhibit the thickest layers, followed by ice clouds and then liquid clouds. The median cloud thickness for these clouds is 2.0XXX, 1.2XXX, and 0.2XXX, respectively. However, due to their considerable variability, no distinct seasonal pattern was discernible for this property. Consequently, thicker layers of mixed-phase clouds contain more ice content than ice clouds layers. Although typical values for both are 0-20 g m^{-2}, mixed phase clouds have systematically higher IWP, with an expressive increase in fall, reaching 20-50 g m^{-2}. Conversely, liquid clouds contain slightly more liquid content than mixed-phase clouds, despite being thin. This indicates that liquid layers consistently maintain thinness, even within mixed-phase clouds. In fact, they have almost the same amount of water, within the interval of 0-20 g m^{-2}, but liquid clouds have higher probability to reach higher values, specially in spring and fall (see figure **??**).

LWC and droplets effective radius does not show a clear seasonal trend for mixed and liquid phase clouds. Effective radius has almost the same distribution along year, with a median of 5 μm for liquid clouds, and is found mainly below 3 km. Differently mixed phase clouds have a radius that can vary a lot from the surface to 10 km, but in general, most frequent values between 5 and 12 μm are found between 1-2 km and 5-9 km. Concerning LWC, despite the fact that it was easy to identify higher values in cold regimes and smaller ones during summer, the variability of data goes through several orders of magnitude. It was observed for liquid and mixed phase clouds.

With respect to ice and mixed phase clouds, the same difficulty to reach a seasonal pattern for IWC was identified. Month profiles did not show a significant difference, except that in summer, values are shifted upwards in the atmosphere. For ice clouds, effective radius can reach 65 μm in summer, but normally they vary between 20-30 μm around 8-12 km. For mixed phase clouds, ice effective radius fluctuates around 50 μm between 1-7 km.

Concerning cloud base height, a significantly seasonal difference on vertical distribution was observed in summer. Normally, the median value of ice cloud base height is 8 kmXXX in fall and 7 kmXXX for the rest of the year. For mixed phase, median values are 4 km and 5 km in cold and warm regimes, respectively. However, an unnusual mode around 5 km appeared in summer, especially for mixed phase clouds, which also present a sharper distribution compared to other seasons. Nonetheless, in this season, liquid clouds presented a spreader distribution, with a median of 3 kmXXX, against a median of 1 kmXXX in other seasons. Cloud top height followed a simmilar behavior. In colder regimes (winter-spring), cloud top is 1.5 kmXXX, 9 kmXXX, 6 kmXXX, for liquid, ice and mixed clouds, respectively. During warm periods (summer-fall), medians changed to 2.5 kmXXX, 8.5 kmXXX, 7.5 kmXXX. However, only in summer, the meadian cloud base height for liquid clouds were 3 kmXXX. Finally, a comprehensive analysis of cloud occurrence and physical properties was conducted at the ACTRIS-CCRESS Granada station. Notably, this is the first study on cloud properties in the southeast of Spain, an area recognized as a hotspot for climate variability. Consequently, it has the potential to enhance our understanding of global cloud properties, as the methods employed here can be applied to other stations as well.

Acknowledgements. TEXT

References

Bühl, J., Seifert, P., Myagkov, A., and Ansmann, A.: Measuring ice-and liquid-water properties in mixed-phase cloud layers at the Leipzig Cloudnet station, Atmospheric Chemistry and Physics, 16, 10 609–10 620, https://doi.org/10.5194/acp-16-10609-2016, 2016.

Hogan, R. J. and O'connor, E. J.: Facilitating cloud radar and lidar algorithms: the Cloudnet Instrument Synergy/Target Categorization product, http://www.mathworks.com, 2004.

Illingworth, A. J., Hogan, R. J., O'Connor, E. J., Bouniol, D., Brooks, M. E., Delanoë, J., Donovan, D. P., Eastment, J. D., Gaussiat, N., Goddard, J. W., Haeffelin, M., Baltinik, H. K., Krasnov, O. A., Pelon, J., Piriou, J. M., Protat, A., Russchenberg, H. W., Seifert, A., Tompkins, A. M., van Zadelhoff, G. J., Vinit, F., Willen, U., Wilson, D. R., and Wrench, C. L.: Cloudnet: Continuous evaluation of cloud profiles in seven operational models using ground-based observations, Bulletin of the American Meteorological Society, 88, 883–898, https://doi.org/10.1175/BAMS-88-6-883, 2007.

Nomokonova, T., Ebell, K., Löhnert, U., Maturilli, M., Ritter, C., and O'Connor, E.: Statistics on clouds and their relation to thermodynamic conditions at Ny-Ålesund using ground-based sensor synergy, Atmospheric Chemistry and Physics, 19, 4105–4126, https://doi.org/10.5194/acp-19-4105-2019, 2019.

Pîrloagă, R., Ene, D., Boldeanu, M., Antonescu, B., O'Connor, E. J., and Ştefan, S.: Ground-Based Measurements of Cloud Properties at the Bucharest–Măgurele Cloudnet Station: First Results, Atmosphere, 13, https://doi.org/10.3390/atmos13091445, 2022.